X-RAY · INSPECTION PIPELINE

Service Design Proposal

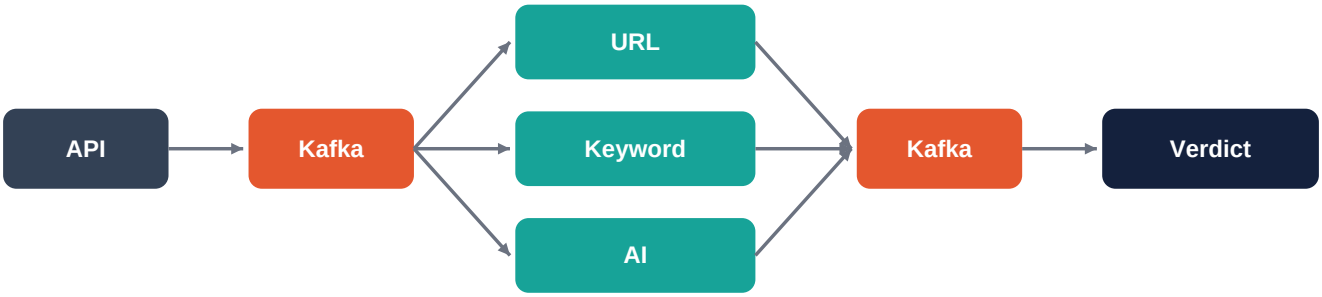
Communication, result contracts, timeouts, retries and identifiers

AT A GLANCE

Topic	Proposal
Internal communication	Kafka between API, engines and Verdict
Inspection result	One common structure; engine data in details
Timeouts	Per-engine time budget within 5s; fail-open on timeout
Retries	At most one quick retry, then mark unavailable
Final verdict	Explainable response: ALLOW / BLOCK / HOLD / FAIL
Identifiers	External msgKey + internal inspection_id + trace_id

01 Internal service communication format

Proposal: use Kafka for the inspection workflow.



independent · parallel

WHY The three engines can work independently and in parallel. Kafka also gives us buffering, decoupling, and independent scaling.

02 Inspection result format

Use **one common result structure** for all engines. Engine-specific information goes inside `details`.

COMMON ENGINE RESULT

```
{
  "msgKey":    "123",
  "traceId":   "abc",
  "engine":    "URL",
  "result":    "BLOCK",
  "reason":    "Malicious domain",
  "details":   {},
  "timestamp": "time format"
}
```

EXAMPLE: AI ENGINE DETAILS

```
"details": {
  "category":    "PHISHING",
  "confidence":  0.92,
  "model":       "model-name / version",
  "explanation": "Short rationale for the decision"
}
```

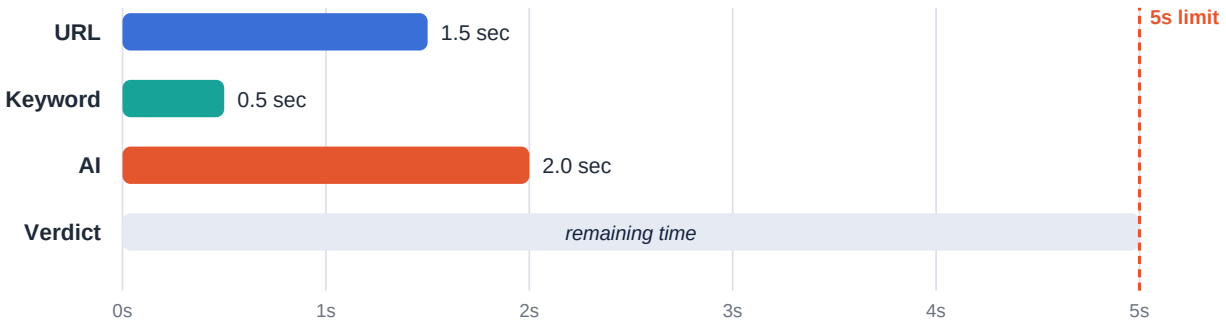
AI details fields above are a placeholder example, to be finalised.

WHY Verdict Service doesn't need to understand completely different message structures from every engine. It can process a common contract.

03 Timeout and partial-result handling

Proposal: give each engine its own time budget within the overall 5-second limit. If an engine doesn't respond in its allotted time, don't wait indefinitely: mark that engine as **timed out** and let Verdict apply the agreed **fail-open** rule.

EXAMPLE BUDGET (TOTAL = 5 SECONDS)

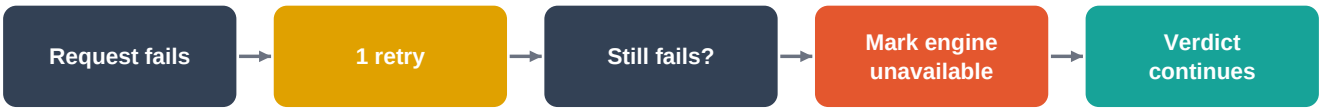


These numbers are only an example; the actual budget is to be agreed with Shahswat.

WHY One slow engine shouldn't hold the entire request indefinitely.

04 Retry strategy

Proposal: keep retries very limited.



WHY If we retry too many times, we can easily violate the 5-second maximum. For transient failures, one quick retry can help. For persistent failures, continuing to retry doesn't help.

05 Final verdict response format

FINAL RESPONSE

```
{
  "msgKey":  "123",
  "verdict": "BLOCK",
  "reason":  "Malicious URL detected",
  "details": {
    "urlResult":     {},
    "keywordResult": {},
    "aiResult":      {}
  }
}
```

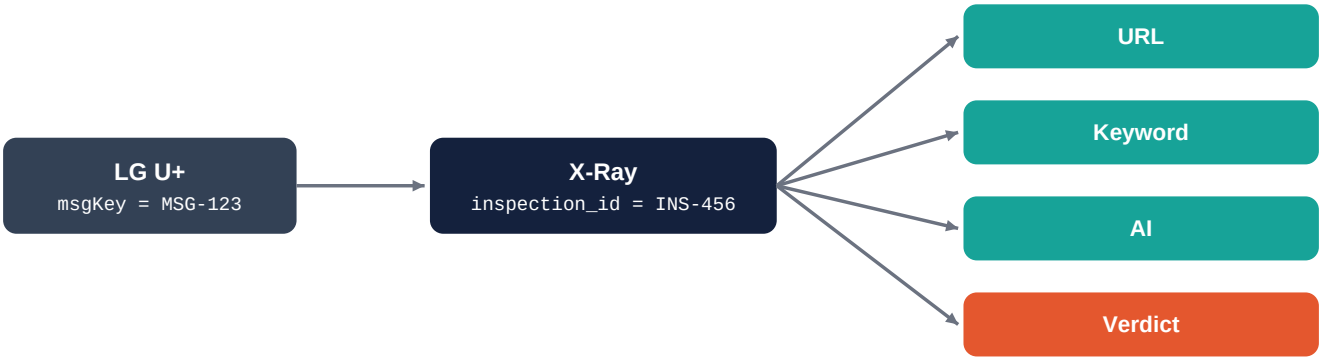
POSSIBLE FINAL RESULTS

ALLOW | BLOCK | HOLD | FAIL

WHY The requirement says the verdict needs to be explainable. Returning only BLOCK isn't enough; we should be able to tell why it was blocked.

06 msgKey vs inspection_id

Keep `msgKey` as the external message identifier and create an internal `inspection_id`.



STORED IDENTIFIERS

Identifier	Example	Owner / purpose
`inspection_id`	`INS-456`	Ours: one X-Ray inspection
`msgKey`	`MSG-123`	Upstream (LG U+) message identifier
`trace_id`	`TRACE-789`	Distributed tracing across services

WHY msgKey belongs to the upstream system, so we shouldn't depend completely on its uniqueness or semantics inside our system. Our own inspection_id gives a clean internal identifier for one X-Ray inspection, and storing all three makes debugging and tracing much easier.